

Complementing nitrogen use efficiency with fertilizer dependency

Quemada, M. and L. Lassaletta

KEY CLAIM

Nitrogen use efficiency (NUE) should be complemented with a decrease in fertilizer dependency if the goal is promoting N inputs other than external fertilizers in cropping systems.

BRIEF JUSTIFICATION, REMAINING CHALLENGES

Three performance indicators are commonly used to provide a perspective of the agroenvironmental sustainability of agricultural systems: N output (or N yield) representing the productivity of the system, N surplus (as the difference between N input and N output) informing about potential environmental risks, and NUE informing about resource efficiency. However, we lack metrics to account for the fertilizer dependency of farms, regions or society as a whole. Promoting such an indicator, calculated as the ratio between human-controllable external inputs and total N inputs, could provide relevant information on whether N fertilizer (either mineral or organic) application can be reduced by increasing direct inputs through biological atmospheric fixation or proper recirculation of already available sources, including reuse of N in other organic products and in water or soil from previous fertilizer application. Additionally, at a regional or national level would allow policymakers to monitor achievements and to evaluate the ability of the system to confront international instability or shocks.

More research is needed to test the applicability of the fertilizer dependency to crop-livestock farms or the whole agro-food system and to assess the N inputs at different spatial and temporal scales.

POTENTIAL FOR SYNERGY WITH OTHER CONTRIBUTIONS

This contribution complements work on farm, regional and national nitrogen budgets, as adding an indicator on fertilizer dependency will provide information on whether N fertilizer application can be reduced by increasing direct inputs through biological atmospheric fixation or proper recirculation of already available sources.

QUESTIONS TO SEED THE DISCUSSION

1. Can we apply an indicator of fertilizer dependency to crop-livestock farms or to the whole agro-food system?
2. Can we improve our information (datasets) on the N inputs (i.e. biological N fixation, atmospheric deposition, recycling,...) at different spatial and temporal scales.

REFERENCES

- Quemada M. and Lassaletta L. 2024. Fertilizer dependency: a new indicator for assessing the sustainability of agrosystems beyond nitrogen use efficiency. *Agronomy for Sustainable Development*. 44 (5): 44.
- Quemada M., Lassaletta L., Jensen L.S., Godinot O., Brentrup F., Buckley C., Foray S., Hvid S.K., Oenema J., Richards K.G., Oenema O. 2020. Exploring nitrogen indicators of farm performance among farm types across several European case studies. *Agricultural Systems*, 177, 102689.
- EU Nitrogen Expert Panel (2015). Nitrogen Use Efficiency (NUE): an indicator for the utilization of nitrogen in agriculture and food systems. Wageningen University, Wageningen, the Netherlands.